LEVEL 9 · PART IV

Helm

9

Goal: Package, version and deploy Kubernetes applications with Helm.

RECOMMENDED TIME	TOPICS
1-2 weeks	6

PRACTICE / CAPSTONE

Create a Helm chart containing deployment, service, ingress and configuration, with per-environment values.

Topics

#	Topic	What you will learn
1	Charts **p.1825**	chart structure, Chart.yaml, helm create, dependencies
2	Values **p.1837**	values.yaml, overrides, -f and --set, schema validation
3	Templates **p.1849**	Go templates, helpers, conditionals, loops, include/tpl, helm template
4	Releases **p.1865**	install/upgrade/rollback/uninstall, release history, hooks, --atomic
5	Repositories **p.1879**	chart repos, OCI registries, helm repo add/search, publishing
6	Packaging Applications for Kubernetes **p.1891**	packaging a real app, lint/test, versioning, umbrella charts
★	Level capstone project **p.1905**	Create a Helm chart containing deployment, service, ingress and configuration, with per-environment values.

How to work through this level

1. Read each topic in order: *what it is* → *why it matters* → *real-world example* → *diagram* → *guide*.
2. Do **every task** in the topic's *Tasks* section and tick the box (`- [x]`) in your own fork or notes.
3. Answer the *Self-check questions* without looking, then reveal the answers.

4. Finish with the **capstone project**, which combines all topics of the level.
5. Move on only when you can do the capstone's *Prove it* task without the guide.

Topic flow

